# Microeconomics I

Class 8 — Choice under uncertainty: expected utility and risk aversion

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slide 1 of 27

## Slide 1 Title

Class 8 opens the second block of the course. Until now the objects of choice were bundles of goods, received for sure; from today they are lotteries, lists of probabilities over prizes. Before the break comes the theory: which preferences over lotteries can be written as an expected utility, and then, with money on the prizes, what it means to dislike risk and how to measure that dislike in euros. After the break, practice on Part I so far. MWG 6.B–6.C.

# Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

quiz page, not numbered

## Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letters of your answers in the block on the sheet, write your name on it, and hand in only the strip.

## Plan for today

| | |
|---|---|
| 10:00 | Short quiz |
| 10:10 | Warm-up; two choices, by hands up |
| 10:17 | Lotteries, continuity, independence |
| 10:28 | Your two choices, revisited |
| 10:31 | Expected utility: form, theorem, uniqueness |
| 10:39 | Money lotteries and risk aversion |
| 10:44 | Certainty equivalent and risk premium |
| 10:50 | Break |
| 11:00 | Practice: five questions on Part I so far |
| 11:44 | Summary |



### Slide 2 Plan for today

Before the break: a warm-up on class 6's true cost of living, then two choices by a show of hands. Then the building blocks: lotteries as probability vectors, compound lotteries, the two new axioms, continuity and independence; the two choices looked at again with the axiom in hand; the expected utility form and its theorem, and why the representation is unique only up to a positive affine transformation; finally money on the prizes: risk aversion, the certainty equivalent and the risk premium. After the break, five practice questions mixing consumer theory (duality; one price rise measured four ways) with today's lotteries.

MWG 6.B–6.C; Muñoz-Garcia chapter 5 is the gentler read.

## Practicalities

- Problem set 2: handed in at the start of class, Wed 21 Oct; returned graded on Tue 27 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu



### Slide 3 Practicalities

Problem set 2 was handed in at the start of class on Wednesday 21 October, on paper, one copy per group; it comes back graded on Tuesday 27 October. Nothing from today's class is on it.

Office hours are on Wednesdays, 14:00–15:00, Room 445, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

## Warm-up: always true, or not?

Only good 1 gets dearer, $p^0 \to p^1$. At $p^0$ she chose $x^0$, with utility $u^0$.

Fixed basket: $p^1 \cdot x^0 \,/\, p^0 \cdot x^0 = 1.3$. True cost-of-living index: $I = e(p^1, u^0) \,/\, e(p^0, u^0)$.

| | |
|---|---|
| $I \leq 1.3$ **true: $x^0$ still reaches $u^0$** | $I \geq 1$ **true: no bundle got cheaper** |
| $I = 1.3$ **false: lower if she substitutes** | good 1 inferior $\Rightarrow I > 1.3$ **false: still $I \leq 1.3$** |

The fixed basket never understates: $1 \leq I \leq 1.3$, normal or inferior.

Class 6, slides 13, 22. Mas-Colell, Whinston and Green (MWG) Props. 3.E.1, 3.E.2, pp. 58–59.



### Slide 4 Warm-up: always true, or not?

The fixed basket prices her old bundle at the new prices; the true index asks what it costs to stay on her old indifference curve. Her old bundle is one way to stay there, so after the rise the cheapest way costs no more than that bundle, $e(p^1, u^0) \leq p^1 \cdot x^0$; before the rise it was the cheapest way, $e(p^0, u^0) = p^0 \cdot x^0$, her whole wealth; dividing, the fixed basket never understates. A dearer good makes no bundle cheaper, so $I$ is at least 1; where it lies in between depends on how easily she replaces good 1 (equal to the fixed basket for perfect complements, $\min\{x_1, x_2\}$; 1.5 against 1.625 for class 6's $x_1 x_2$ consumer).

Nothing in this argument uses a wealth effect, so normal and inferior goods obey the same bounds. Normal versus inferior decides a different comparison, from class 7: whether the equivalent variation is above or below the compensating variation.

## Two choices: which would you take?

Imagine the prizes are real and paid once. No right answer.

**First choice**

**A**: 1 million euros for sure

**B**: 10 %: 5 million euros
89 %: 1 million euros
1 %: nothing

**Second choice**

**C**: 11 %: 1 million euros
89 %: nothing

**D**: 10 %: 5 million euros
90 %: nothing

**Hands up: A or B? Then: C or D?**

Hypothetical prizes; the structure of Allais (1953), as in MWG Example 6.B.2, p. 179.



### Slide 5 Two choices: which would you take?

Two pairs of hypothetical prizes, decided by a show of hands before any theory. There is no right answer: each choice on its own is a matter of taste, and the only thing the theory has to say is about the two choices taken together. The counts were taken in class and discussed on slide 11, right after the independence axiom; they are not printed here.

The structure is Allais (1953), as in MWG Example 6.B.2, p. 179. The key is what A has in common with B, and C with D, prize by prize; slide 11 writes it out.

# Lotteries and expected utility

MWG 6.B



## A simple lottery is a list of probabilities: a point in a triangle

Simple lottery: $L = (p_1, \ldots, p_N)$, $p_n \geq 0$, $\sum_n p_n = 1$; $p_n$ = probability of outcome $n$.

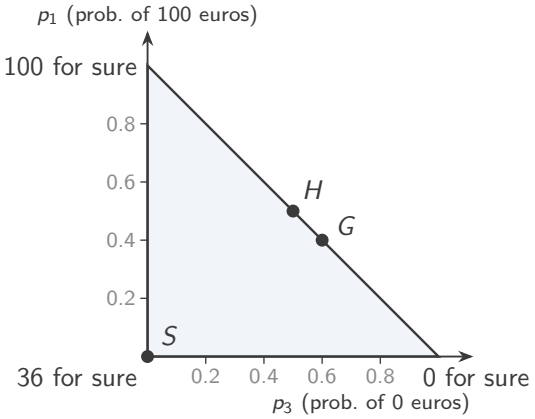
Outcomes: 100, 36, 0 euros.

$S = (0, 1, 0)$: 36 euros for sure

$H = (0.5, 0, 0.5)$: a coin, 100 or 0

$G = (0.4, 0, 0.6)$: 40 % chance of 100

MWG Def. 6.B.1, p. 168; Fig. 6.B.1, p. 169 draws the same set as an equilateral triangle.




$p_2 = 1 - p_1 - p_3$: 0 on the hypotenuse, 1 at $S$



**Slide 6** Lotteries and expected utility

Part 1: lotteries, the axioms, and the expected utility form. MWG 6.B.

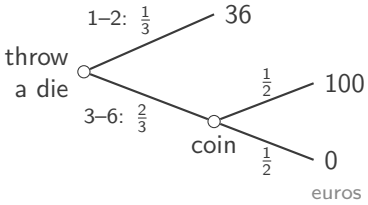
**Slide 7** A simple lottery is a list of probabilities: a point in a triangle

A lottery is described by its probabilities, not by its prizes: the prizes are fixed once and for all (here 100, 36 and 0 euros, in this order), and a lottery is the vector that says how likely each one is. The entries of the vector are probabilities, so they lie between 0 and 1 and add up to 1; a vector of prizes is not a lottery, which is the first notation trap of the block.

With three prizes the third probability is fixed by the other two, so every lottery is a point of a triangle: the corners are the sure prizes, each side holds the lotteries that never pay one of the prizes, and the hypotenuse here holds those that never pay 36. The three named lotteries $S$, $H$ and $G$ are used all morning. MWG draws the same set as an equilateral triangle (Fig. 6.B.1, p. 169); flattening it changes nothing.

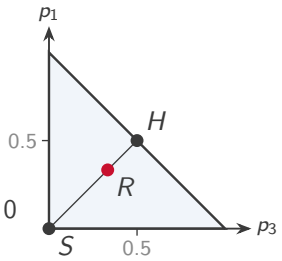
## A compound lottery reduces to a simple one: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Compound lottery $(L_1, \ldots, L_K; \alpha_1, \ldots, \alpha_K)$: you get $L_k$ with probability $\alpha_k$.
Reduced lottery: $\alpha_1 L_1 + \cdots + \alpha_K L_K$.




$R = (S, H;\ \frac{1}{3}, \frac{2}{3})$

$\frac{1}{3}(0,1,0) + \frac{2}{3}(0.5,0,0.5)$
$= (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
= one throw: 1–2 → 100,
3–4 → 36, 5–6 → 0




Only the reduced lottery matters (MWG's consequentialist premise): the die-then-coin tree and one throw are the same lottery.

MWG Def. 6.B.2, pp. 169–170; Figs. 6.B.2–6.B.3, p. 170.



## Rational and continuous preferences over lotteries

Rational: complete, transitive (class 1).

Continuous: for all $L, L', L'' \in \mathcal{L}$ the sets $\{\alpha \in [0,1] : \alpha L + (1-\alpha)L' \succsim L''\}$ and $\{\alpha \in [0,1] : L'' \succsim \alpha L + (1-\alpha)L'\}$ are closed.

Small probability changes never reverse a strict ranking.

> **Fails: safety first** (no chance of 0 beats any chance).
> $L = 100$, $L' = 0$, $L'' = 36$ euros for sure: second set $= [0, 1)$.

MWG Def. 6.B.3, p. 171 ("safety first": same page). Class 3: lexicographic order, not continuous.



**Slide 8** A compound lottery reduces to a simple one: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

A compound lottery is a lottery whose prizes are themselves lotteries. The rule for reducing it follows from how probabilities combine along a path: multiply along the branches, then add over the paths that end in the same prize. Here the coin is reached only two times in three, which is why the 100 and the 0 each get $\frac{2}{3} \cdot \frac{1}{2}$ and not $\frac{1}{2}$; adding probabilities across the stages is the slip to avoid.

In vectors the reduced lottery is the weighted average of the component lotteries, $\frac{1}{3}S + \frac{2}{3}H$, which in the triangle is the point two thirds of the way from $S$ to $H$. Every result that follows uses this linear structure. The premise of the whole theory (MWG p. 170) is that only the reduced lottery matters: two trees with the same reduction are the same object of choice.

**Slide 9** Rational and continuous preferences over lotteries

Rationality is the same pair of axioms as in class 1, now on the set of lotteries; MWG notes that it asks more here, because comparing risky alternatives is harder than comparing bundles (p. 171). Continuity is written with mixtures: the sets of weights $\alpha$ for which the mix is at least as good as a third lottery, or at most as good, must be closed. In words, a small change in the probabilities never turns a strict preference around; MWG's example is a car trip with a tiny chance of an accident, still better than staying at home.

The counterexample is class 3's lexicographic idea in lottery form: a consumer who puts safety first ranks any lottery with no chance of 0 above any lottery with one. The card names the instance that fails: with $L$ and $L'$ the sure 100 and the sure 0, and $L''$ the sure 36, the set of weights for which 36 is at least as good is $[0, 1)$, which misses its limit point. As in class 3, continuity is what makes a utility function on lotteries exist at all; independence, next, is what gives it a formula. The two axioms are easy to mix up: continuity is about limits, independence about mixing with a common third lottery.

## Independence: mixing both with the same $L''$ keeps their ranking

Independence axiom: for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0,1)$:

$$L \succsim L' \iff \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$




$\alpha L + (1-\alpha)L''$ $\qquad$ $\alpha L' + (1-\alpha)L''$

Tails gives both the same $L''$, and you get it instead of $L$ or $L'$, never with it: unlike bundles of goods.

MWG Def. 6.B.4, p. 171 (fn. 5: von Neumann and Morgenstern, 1944); coin-toss reading and Fig. 6.B.4, p. 172.



**Slide 10** Independence: mixing both with the same $L''$ keeps their ranking

The axiom that makes the theory: a ranking of two lotteries survives when both are mixed, with the same weight, with the same third lottery. The coin picture explains why one would accept it: on tails both coins pay the same $L''$, so nothing differs; on heads one coin pays $L$ and the other $L'$; so the coin with the preferred lottery should be preferred, whatever $L''$ is.

Why this is a genuinely new assumption, and not class 1 again, is in MWG's remark on p. 172: with goods, how much you like apples can depend on the pears you also get, because you consume them together; here $L''$ comes instead of $L$, never with it. Read the quantifiers: the axiom asks for every $\alpha$ strictly between 0 and 1 and every $L''$. It also holds with strict preference and with indifference in place of the weak ranking (MWG Exercise 6.B.1, p. 172), which the next slide and Practice 1 use.

## A with D breaks independence: the shared 89 % should not matter

| | 11 % of the time | 89 % of the time |
|---|---|---|
| **A** | 1 million euros | 1 million euros |
| **B** | $L'$ | 1 million euros |
| **C** | 1 million euros | 0 |
| **D** | $L'$ | 0 |

$L'$: $\frac{10}{11}$ chance of 5 million euros, $\frac{1}{11}$ chance of 0 euros.

Independence with $\alpha = 0.11$: A $\succsim$ B $\iff$ 1 million for sure $\succsim L' \iff$ C $\succsim$ D.

Strictly preferring A and D breaks it.

Allais (1953); MWG Example 6.B.2, pp. 179–180, and fn. 8: "roughly half the students choose this way". 81 experiments: "a fragile empirical finding" (Blavatskyy, Ortmann & Panchenko 2022, American Economic Journal: Microeconomics 14(1)).



**Slide 11** A with D breaks independence: the shared 89 % should not matter

The two choices of slide 5, rewritten over 100 tickets. On tickets 12 to 100 (89 % of the draws) A and B pay the same prize, and so do C and D: the common consequence, 1 million in the first pair and nothing in the second. On tickets 1 to 11 the first pair and the second pair offer exactly the same two alternatives: 1 million for sure, or the lottery $L'$ with ten chances in eleven of 5 million. So the first choice and the second differ only in the shared column, and that is slide 10's axiom with $\alpha = 0.11$: changing the common $L''$ cannot change the ranking, so the consistent answers are A with C, or B with D.

Strictly preferring A and D is not intransitive and not irrational on its own: each choice alone is fine, and what expected utility forbids is the pair; MWG reports that roughly half of their students choose it (fn. 8, p. 179). In expected utility algebra the two choices become the same inequality after adding the same term to both sides (MWG p. 180). What to make of it is open: a mistake one corrects on reflection, or a reason to want theories without independence; those are outside this course. The slide's second source calls the pattern, across 81 experiments, "a fragile empirical finding".

## Expected utility: $U(L) = u_1 p_1 + \cdots + u_N p_N$ is linear in probabilities

$U$ has the expected utility form if there are numbers $u_1, \ldots, u_N$ with $U(L) = u_1 p_1 + \cdots + u_N p_N$ for every $L$.

Such a $U$ is a von Neumann–Morgenstern (vNM) expected utility function.

Her utilities for 100, 36, 0 euros: 10, 6, 0.

$U(S) = 6, \quad U(H) = 5$

$U(R) = \frac{1}{3} \cdot 10 + \frac{1}{3} \cdot 6 + \frac{1}{3} \cdot 0 = \frac{16}{3} = \frac{1}{3}\, U(S) + \frac{2}{3}\, U(H)$

MWG Def. 6.B.5 and Prop. 6.B.1 (expected utility form $\iff$ $U$ linear), p. 173.



**Slide 12** Expected utility: $U(L) = u_1 p_1 + \cdots + u_N p_N$ is linear in probabilities

The expected utility form attaches one number $u_n$ to each prize and values a lottery by the probability-weighted average of those numbers: the same operation as the expected value $E[x] = \sum_n p_n x_n$, with utilities in place of euros. The notation trap is the pairing: each probability multiplies the utility of its own prize, as in the dot product $p \cdot u$.

Her utilities 10, 6, 0 for 100, 36 and 0 euros give the first sign of risk aversion: she prefers 36 for sure to a fair coin for 100 or 0, although the coin's expected prize is 50 euros. The die lottery $R$ of slide 8 shows the property that matters: its utility is the weighted average of the utilities of its parts. That is MWG Prop. 6.B.1, p. 173: having the expected utility form is the same as being linear in the probabilities.

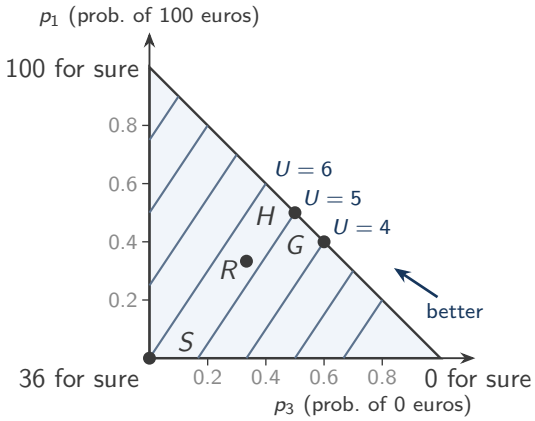
## Continuity + independence ⇒ expected utility: straight, parallel lines

Expected utility theorem: if $\succsim$ on $\mathcal{L}$ is rational, continuous and satisfies independence, there are $u_1, \ldots, u_N$ such that for all $L, L'$:
$L \succsim L' \iff \sum_n u_n p_n \geq \sum_n u_n p'_n$.

In the triangle: indifference curves are straight, parallel lines.

Conversely, every expected utility $U$ satisfies independence (MWG Ex. 6.B.2).

MWG Prop. 6.B.3, (6.B.4), p. 176; proof pp. 176–178; Fig. 6.B.5, p. 175.






**Slide 13** Continuity + independence ⇒ expected utility: straight, parallel lines

The theorem says the three axioms are exactly what an expected utility representation needs: rational, continuous and independent preferences over lotteries can be written as $\sum_n u_n p_n$, and (this slide's last line) any such $U$ satisfies the axioms. The proof idea is short. By continuity every lottery is exactly as good as some mix of the best prize and the worst, and the weight on the best prize can serve as its utility; independence then forces this utility to be linear in the probabilities. The details are in MWG pp. 176–178.

The picture is the test to remember: a linear $U$ has level sets that are straight lines with one common slope, so the indifference map of an expected utility maximizer in the triangle is a family of straight, parallel lines, here of slope 1.5. Curved or fanning indifference lines are not a different taste; they contradict independence (MWG Fig. 6.B.5, p. 175). The trap is to draw them bowed, like indifference curves over goods.

## Expected utility is unique up to $\beta U + \gamma$, $\beta > 0$; $u^2$ flips her choice

| utilities of (100, 36, 0 euros) | $U(G)$ $G = 40\,\%$ of 100 | $U(S)$ $S = 36$ for sure | she picks |
|---|---|---|---|
| $u = (10, 6, 0)$ | 4 | 6 | $S$ |
| $2u + 3 = (23, 15, 3)$ | 11 | 15 | $S$ |
| $u^2 = (100, 36, 0)$ | 40 | 36 | $G$ |

$\tilde{U}$ is another vNM utility for $\succsim \iff \tilde{U} = \beta U + \gamma$ with $\beta > 0$.

$u^2$ keeps the order of prizes, not lotteries:
unlike class 3, expected utility is cardinal.

MWG Prop. 6.B.2, pp. 173–174.



**Slide 14** Expected utility is unique up to $\beta U + \gamma$, $\beta > 0$; $u^2$ flips her choice

Class 3 said that utility over bundles is ordinal: any increasing transformation represents the same preferences. Over lotteries, an increasing transformation of $U$ still ranks lotteries alike but is no longer an expected utility; an increasing transformation of the prize utilities $u_n$ can reverse a choice. Doubling and adding 3 rescales every lottery's value alike, so the choice between $G$ and $S$ stays; squaring the prize utilities keeps the sure prizes in order yet reverses the choice, because an average of squares is not the square of an average. So an expected utility function is unique up to $\beta U + \gamma$ with $\beta > 0$ (MWG Prop. 6.B.2, pp. 173–174): it is cardinal.

The squared person's utilities are 100, 36 and 0, the prizes themselves: this consumer ranks lotteries by expected money and returns on slide 16. The trap is the class 3 habit of applying any increasing $f$ to the $u_n$: only $\beta U + \gamma$ keeps both the ranking and the expected utility form.

# Money lotteries and risk aversion

MWG 6.C



**Slide 15** Money lotteries and risk aversion

Part 2: money lotteries, risk aversion, the certainty equivalent and the risk premium. MWG 6.C.

## A money lottery is a distribution function $F$; $U(F) = \int u(x)\,dF(x)$

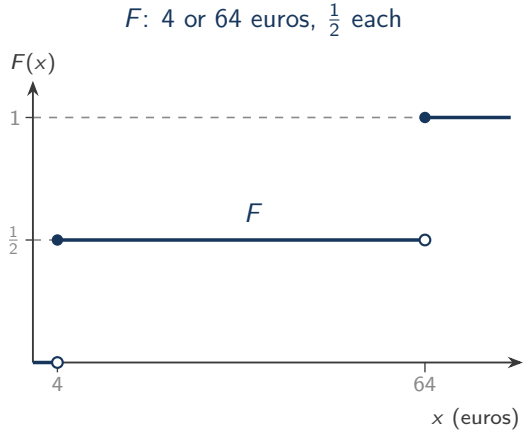
$F(x)$ = probability that the payoff is at most $x$.

$U(F) = \int u(x)\,dF(x)$; with prizes $x_n$ and probabilities $p_n$: $\sum_n p_n\,u(x_n)$.

$U$: vNM utility, on lotteries.
$u$: Bernoulli utility, on money — increasing, continuous.

Her 10, 6, 0 for 100, 36, 0 euros: $u(x) = \sqrt{x}$.

MWG p. 183; (6.C.1) and fn. 12, p. 184; p. 185.






**Slide 16** A money lottery is a distribution function $F$; $U(F) = \int u(x)\,dF(x)$

With money prizes a lottery can have any number of outcomes, so it is described by its distribution function, the probability of receiving at most $x$; for the lottery on the slide $F$ is 0 below 4 euros, jumps by a half at 4 and again at 64. Expected utility is the same formula as on slide 12, written as an integral; with finitely many prizes it is the sum $\sum_n p_n u(x_n)$, the expected value of $u(x)$.

Two functions carry the name utility, and the course keeps them apart as MWG does (fn. 12, p. 184): $U$, the von Neumann–Morgenstern utility, takes a lottery; $u$, the Bernoulli utility, takes an amount of money, and it is increasing and continuous. Calling $u$ the vNM utility is the naming slip to avoid. Her numbers from the first hour were $\sqrt{x}$ at 100, 36 and 0 all along, and the squared person of slide 14 has $u(x) = x$; both return on the next slides.

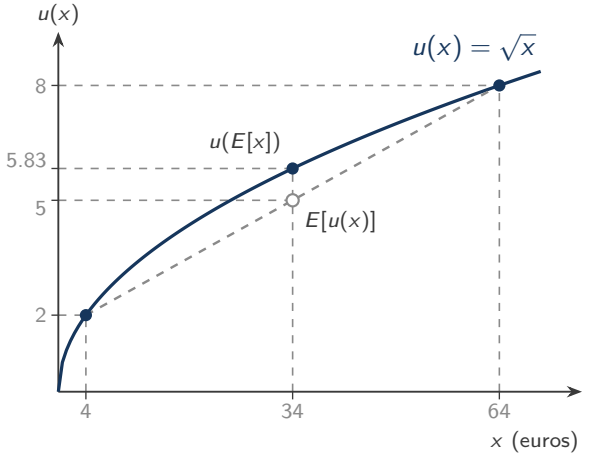
## Risk averse ⇔ $u$ concave: the chord lies below the graph

Risk averse: for every $F$, the sure amount $\int x\,dF(x)$ is at least as good as $F$.

With expected utility: $\int u(x)\,dF(x) \le u(\int x\,dF(x))$ for every $F$ — Jensen: $u$ concave.

Concave in money: a property of the function $u$, not convexity of preferences over bundles (classes 2–3). $U$ is linear in probabilities.

MWG Def. 6.C.1 and (6.C.2), p. 185; Fig. 6.C.2, p. 186.






**Slide 17** Risk averse ⇔ $u$ concave: the chord lies below the graph

Risk aversion is defined without any utility function: for every lottery, its expected value paid for sure is at least as good as the lottery itself (MWG Def. 6.C.1, p. 185); risk neutral means always indifferent, strictly risk averse means the sure amount is strictly better whenever the lottery is not degenerate. For an expected utility maximizer the definition becomes Jensen's inequality, the average of $u$ at most $u$ at the average, for every $F$, and that is exactly concavity of $u$: the chord lies below the graph. The figure reads it on slide 16's lottery: the chord's midpoint is $E[u(x)] = 5$, the curve at $E[x] = 34$ is about 5.83.

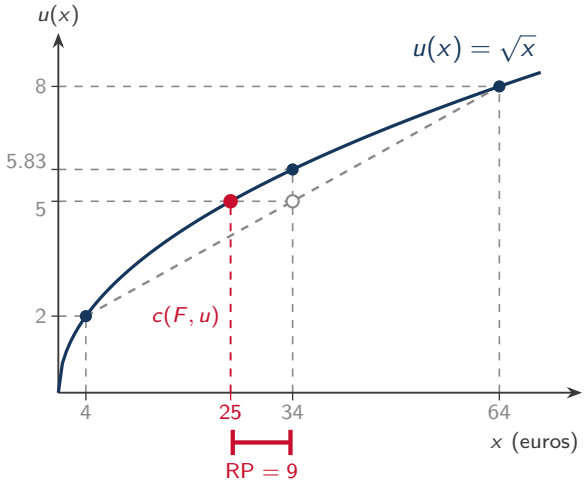
Two re-anchorings from classes 2 and 3 matter here. This is concavity of a function of money, checked by the chord test; it is not convexity of preferences over bundles and not quasiconcavity, and $U$ itself is always linear in the probabilities, so risk aversion is never about curvature in $p$. The boundary trap of the block is to write $u(E[x])$ for $E[u(x)]$: the first is a point on the curve, the second a point on the chord, and they are equal only when $u$ is linear.

## Certainty equivalent 25 euros, risk premium 9 euros: three steps

Certainty equivalent $c(F,u)$:
$u\big(c(F,u)\big) = \int u(x)\,dF(x)$.
Risk premium: $\mathrm{RP} = E[x] - c(F,u)$.

1. $E[x] = \frac{1}{2}\cdot 4 + \frac{1}{2}\cdot 64 = 34$ euros
2. $E[u(x)] = \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 8 = 5$ utils
3. $c(F,u) = u^{-1}(5) = 25$ euros;
   $\mathrm{RP} = 34 - 25 = 9$ euros

MWG Def. 6.C.2(i), (6.C.3), p. 186: "some expected return is traded for certainty". RP as in Muñoz-Garcia, p. 357.

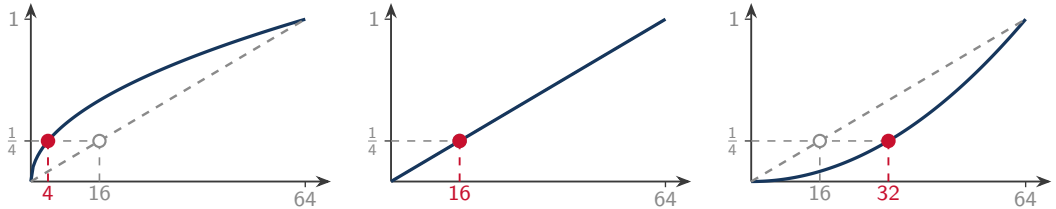





## Concave $u$: $c(F,u) \le E[x]$ (MWG Prop. 6.C.1); convex $u$: $c(F,u) \ge E[x]$

Toss two coins. Both heads: 64 euros; otherwise nothing. $E[x] = 16$ euros for everyone.

Ana: $u = \sqrt{x}/8$ Ben: $u = x/64$ Carla: $u = x^2/4096$




$c = 4 < 16$ $c = 16$ $c = 32 > 16$

Each $u$ rescaled so that $u(0) = 0$, $u(64) = 1$ (Prop. 6.B.2): $E[u(x)] = \frac{1}{4}$ in all three.

Risk averse $\iff$ $u$ concave $\iff$ $c(F,u) \le E[x]$ for every $F$, i.e. RP $\ge 0$.

MWG Prop. 6.C.1, p. 187.



**Slide 18** Certainty equivalent 25 euros, risk premium 9 euros: three steps

The certainty equivalent is the sure amount she finds exactly as good as the lottery, defined by $u(c) = E[u(x)]$ (MWG Def. 6.C.2(i), p. 186), and the risk premium is the expected value she gives up to be rid of the risk, $E[x] - c(F,u)$. The recipe never changes: the expected value, in euros; the expected utility, in utils; then back to euros through $u^{-1}$. Here 5 utils correspond to 25 euros, so she would swap a lottery worth 34 euros on average for 25 sure euros, a premium of 9 euros.

The three slips all cross a boundary between the steps: using $u(E[x])$, about 5.83, in place of $E[u(x)] = 5$; reporting the certainty equivalent as 5, a number of utils, instead of 25 euros; and flipping the sign of the premium. MWG does not name the risk premium and only says that some expected return is traded for certainty; Muñoz-Garcia (p. 357) names it, as $E[x] - c(F,u)$, and the equivalent form $u(E[x] - \mathrm{RP}) = E[u(x)]$ says the same thing. In the picture the certainty equivalent is where the curve reaches the chord's height, to the left of the mean for a concave $u$.

**Slide 19** Concave $u$: $c(F,u) \le E[x]$ (MWG Prop. 6.C.1); convex $u$: $c(F,u) \ge E[x]$

Toss two coins: both heads pay 64 euros, otherwise nothing, so the expected prize is 16 euros whoever plays. Rescaling each Bernoulli utility ($\sqrt{x}$, $x$, $x^2$) by a positive affine transformation, allowed by slide 14, so that $u(0) = 0$ and $u(64) = 1$ puts all three in one box: the same chord, and an expected utility of a quarter for everyone. The certainty equivalent is where each curve reaches that height: 4 euros for the concave curve, 16 for the straight line, 32 for the convex one. That is MWG Prop. 6.C.1, p. 187: risk aversion, concavity of $u$ and a certainty equivalent never above the expected value are one property, because $u$ is increasing.

Risk averse does not mean refusing every bet, only fair ones: a risk averter with a differentiable $u$ takes a small enough share of any bet with a positive expected gain (MWG p. 189). Who is more risk averse than whom, and how to measure it, opens Tuesday 27 October.

# Break

Back at 11:00

slide 20 of 27

## Slide 20 Break

Ten minutes.

# Practice

Five questions on Part I so far: try first, then we solve them together

slide 21 of 27

## Slide 21 Practice

The second half is practice on Part I so far: consumer theory and today's lotteries. Try each question on your own first; then it is solved at the board. The questions mix classes, so the first job in each is to recognise which idea it rests on.

## Practice 1: two dice games

Prizes 150, 60, 0 euros; a lottery lists their probabilities in this order.

Iris's preferences over lotteries satisfy the independence axiom. Throw a fair die:

**Game X**: on 1–3, 60 euros for sure; on 4–6, the lottery $M = (0, \frac{1}{2}, \frac{1}{2})$.

**Game Y**: on 1–3, a fair coin for 150 or 0 euros; on 4–6, $M$.

Iris strictly prefers Y to X.

(a) Reduce X and Y.

(b) Which must be true?

A She strictly prefers 60 euros for sure to the coin.

B She is risk loving.

C She strictly prefers the coin to 60 euros for sure.

D Nothing follows unless we know how much she values 150, 60 and 0 euros.



## Practice 2: three equally likely prizes

A lottery $F$ pays 0, 9 or 36 euros, each with probability $\frac{1}{3}$.

Rosa is an expected utility maximizer with Bernoulli utility $u(x) = \sqrt{x}$.

(a) $E[x]$ and $E[u(x)]$.

(b) Her certainty equivalent $c(F, u)$ and the risk premium $\text{RP} = E[x] - c(F, u)$.

(c) Instead of $F$ she is offered 12 euros for sure. Does she accept? Would a risk-neutral person accept?



### Slide 22 Practice 1: two dice games

Two things from today in one question: slide 8's rule for reducing a compound lottery and slide 10's independence axiom. In (b) the direction is new: the ranking of the two games is given, and the question is what it says about the branches on which they differ; both games pay $M$ on three of the six faces, and the axiom, written as an equivalence, reads in both directions.

In (a) the slip is to add the die's probabilities to the lotteries' instead of multiplying along the branches. In (b), compare the sure 60 euros with the coin's expected value before reading any attitude to risk into the choice. The answers are on the slides released after class.

### Slide 23 Practice 2: three equally likely prizes

A straight run through slide 18's three steps: the expected value in euros, the expected utility in utils, then back to euros through the inverse of $u$; the risk premium (RP) is the expected value minus the certainty equivalent. The prizes were chosen so that the square roots come out whole.

The slip is the boundary between the steps: a number of utils is never a certainty equivalent, and $u$ at the expected prize is not the expected utility. For (c), set the sure offer against the right one of your two euro amounts, then ask which amount a risk-neutral person would use instead. The answers are on the slides released after class.

## Practice 3: what duality says

Vera's preferences are rational, continuous, locally non-satiated and strictly convex. At prices $p = (2, 7)$, the cheapest bundle that gives her utility 5 is $h(p, 5) = (7, 4)$.

$h$: Hicksian demand; $x$: Walrasian demand.

**Which statement must be true?**

A $h((4, 14),\ 5) = (14, 8)$

B $x(p, 42) = (7, 4)$

C $x(p, 84) = (14, 8)$

D $x(p, 40) = (7, 4)$



## Practice 4: one price rise, four measures

Kim: $u(x_1, x_2) = x_1 x_2$, wealth $w = 18$.
Good 1 gets dearer: prices $p^0 = (1, 3) \to p^1 = (9, 3)$.

Class 4: $x(p, w) = \big(w/(2p_1),\ w/(2p_2)\big)$. $h$: Hicksian demand; $e$: expenditure function.

(a) $x^0 = x(p^0, w)$, $u^0$; $x^1 = x(p^1, w)$, $u^1$.

(b) The Hicksian bundle $h(p^1, u^0)$; on good 1, the substitution effect $h_1(p^1, u^0) - x_1^0$ and the wealth effect $x_1^1 - h_1(p^1, u^0)$.

(c) Compensating variation
$\text{CV} = w - e(p^1, u^0)$ and equivalent variation
$\text{EV} = e(p^0, u^1) - w$.

(d) The true cost-of-living index $e(p^1, u^0)/w$ and the fixed basket $(p^1 \cdot x^0)/(p^0 \cdot x^0)$.



### Slide 24 Practice 3: what duality says

What duality says and what it does not, this time starting from the Hicksian side. The question rests on class 6's link between the expenditure problem and the utility problem (MWG Prop. 3.E.1): the cheapest bundle for a given utility level is also the best bundle at some wealth, and the proposition says which wealth. Each option moves one of the two demands to new prices or to new wealth.

The trap is to assume that doubling an argument of $h$ or $x$ doubles the bundle; ask instead what each is homogeneous in, from classes 4 and 6, and what the bundle costs at the prices given. The answers are on the slides released after class.

### Slide 25 Practice 4: one price rise, four measures

One price rise measured four ways. The exercise joins class 4's demand formula, class 6's Hicksian demand, expenditure function and two cost-of-living indices (reused in slide 4's warm-up), and class 7's definitions of the compensating variation (CV) and the equivalent variation (EV). Part (b) is the split into substitution and wealth effects: first the bundle that keeps her old utility at the new prices, then the one she actually buys.

In (c), decide before computing whether a price rise should make each variation positive or negative with the slide's definitions, then check your numbers against that. Part (d) puts numbers on the warm-up's claims. The answers are on the slides released after class.

## Practice 5: a straight-line Bernoulli utility

Otto is an expected utility maximizer with Bernoulli utility $u(x) = 3x + 4$ over money $x \geq 0$, in euros.

**Which statement is true?**

A He is risk averse, since $u(0) = 4 > 0$.

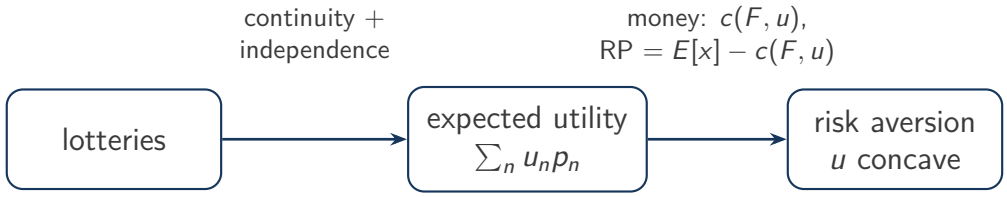
B His certainty equivalent of every lottery is 4 euros above its expected value.

C He strictly prefers 10 euros for sure to a fair coin for 0 or 20 euros.

D His certainty equivalent of every lottery equals its expected value.



## Today in one line




Tuesday 27 October: Uncertainty II: Arrow–Pratt measures, comparing risk aversion, insurance. First- and second-order stochastic dominance, mean-preserving spreads. Review.

Read MWG 6.C–6.D (or Muñoz-Garcia ch. 5) for Tuesday 27 October.

**Part I exam: Wednesday 28 October, Room 1030.**
It runs from 09:30 to 11:20.



### Slide 26 Practice 5: a straight-line Bernoulli utility

A test of slide 17's result that an expected utility maximizer's attitude to risk is read from the shape of the Bernoulli utility, together with slide 14's point that a positive affine transformation of $u$ leaves the ranking of lotteries alone. Each option is a claim about a straight-line $u$.

The trap is to read the intercept or the slope as if they carried information about risk. Decide first what shape $u$ has and what slide 17 says about that shape; then check each option against it, the one with the fair coin by slide 18's three steps. The answers are on the slides released after class.

### Slide 27 Today in one line

In one line: a lottery is a vector of probabilities, and a compound lottery is the weighted average of its parts; with continuity and independence, preferences over lotteries are an expected utility, straight and parallel in the triangle, and unique up to a positive affine transformation, so that expected utility is cardinal; with money prizes, risk aversion is concavity of the Bernoulli utility, and the certainty equivalent and the risk premium measure it in euros through the three steps.

Tuesday 27 October: Uncertainty II: Arrow–Pratt measures, comparing risk aversion, insurance; first- and second-order stochastic dominance, mean-preserving spreads; review. Reading: MWG 6.C–6.D, or Muñoz-Garcia chapter 5.

Part I exam: Wednesday 28 October, Room 1030. It runs from 09:30 to 11:20.